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SUBSTRATE WARPAGE COMPENSATION

Abstract

Embodiments of the disclosure include apparatus and methods for substrate processing. A method includes applying a first DC bias to a first pair of electrodes and a second pair of electrodes, the first pair is disposed within a first portion of a substrate support and the second pair is disposed within a second portion of the substrate support. The method further includes detecting a difference in a first electrical characteristic of a first flow and a second electrical characteristic of a second flow between the electrode pairs and the substrate. A second DC bias is applied to the first pair and a third DC bias, different from the second DC bias, is applied to the second pair. The second DC bias and the third DC bias are selected based on the difference in the first and second flows.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. Provisional Patent Application Ser. No. 63/560,362, filed Mar. 1, 2024, which is incorporated herein by reference.

BACKGROUND
Field
[0002] Embodiments described herein generally relate to a system and methods used in semiconductor device manufacturing. More specifically, embodiments of the present disclosure relate to compensating for substrate warpage that occurs during manufacturing processes.
Description of the Related Art
[0003] In semiconductor and device packaging manufacturing applications, multiple substrates, "wafers" or device packages are disposed over a surface of a substrate support for processing. Ensuring a flatness of the substrates, "wafers" or device packages on the substrate support surface during the various manufacturing processes generally improves the quality, reliability, and yield of manufactured semiconductor devices and device packages that can include 3D chip stacking configurations. However, maintaining the flatness of the substrates or die packages on the surface of the substrate support is challenging because significant variations in the deposited films' film stress or package interconnect layouts and/or high processing temperatures during the various manufacturing processes affect stresses formed in substrate materials causing the substrates to warp/bow. The warped substrates or device packages are no longer flat on the surface of the substrate support, and suffer from wide temperature variations which affect the quality/yield of the devices formed on the substrate or device packages.
[0004] Accordingly, there is a need in the art for a desirable substrate processing system that solves the problems described above.
SUMMARY
[0005] To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the appended drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed.
[0006] Embodiments of the present disclosure provide a method for substrate processing. The method generally includes chucking a substrate to a surface of a substrate support that is disposed in a processing volume by applying a first DC bias to a first pair of electrodes and a second pair of electrodes. The first pair of electrodes is disposed within a first portion of the substrate support and the second pair of electrodes is disposed within a second portion of the substrate support. An AC bias is applied to the first pair of electrodes and the second pair of electrodes. A difference is detected in a first electrical characteristic formed between the first pair of electrodes and the substrate and a second electrical characteristic formed between the second pair of electrodes and the substrate. A second DC bias is applied to the first pair of electrodes and a third DC bias is applied to the second pair of electrodes. The second DC bias is different from the third DC bias, and the second DC bias and the third DC bias are selected based on detecting the difference in the first electrical characteristic and the second electrical characteristic.
[0007] Embodiments of the present disclosure provide an apparatus which includes a substrate support and a non-transitory computer readable medium storing executable instructions that, when executed by at least one processor, cause the at least one processor to perform operations for chucking a substrate to a surface of the substrate support. The operations include applying a first DC bias to a first pair of electrodes and a second pair of electrodes. The first pair of electrodes is

disposed within a first portion of the substrate support and the second pair of electrodes is disposed within a second portion of the substrate support. An AC bias is applied to the first pair of electrodes and the second pair of electrodes. A difference is detected in a first AC current flowing between the first pair of electrodes and a second AC current flowing between the second pair of electrodes. A second DC bias is applied to the first pair of electrodes and a third DC bias is applied to the second pair of electrodes. The second DC bias is different from the third DC bias, and the second DC bias and the third DC bias are selected based on detecting the difference in the first AC current and the second AC current.

[0008] Embodiments of the present disclosure provide an electrostatic chuck (ESC) which includes a substrate support and a plurality of pairs of bipolar electrodes that each have interdigitated conductors. A first plurality of bipolar electrodes of the plurality of pairs of bipolar electrodes is included in a first group of bipolar electrodes and a second plurality of bipolar electrodes of the plurality of pairs of bipolar electrodes is included in a second group of bipolar electrodes. A plurality of chucking regions are formed in a surface of the substrate support. At least one pair of bipolar electrodes in the first plurality of bipolar electrodes and at least one pair of bipolar electrodes in the second plurality of bipolar electrodes are disposed within each of the plurality of chucking regions. The first group of bipolar electrodes are configured to be separately biasable relative to the second group of bipolar electrodes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] So that the manner in which the above recited features of embodiments of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

[0010] FIG. **1**A is a schematic representation of an example substrate processing system, in accordance with certain embodiments of the present disclosure.

[0011] FIG. **1**B illustrates an example of substrate warpage occurring during substrate processing, in accordance with certain embodiments of the present disclosure.

[0012] FIG. **1**C illustrates example arrangements of pairs of electrodes embedded in an electrostatic chuck (ESC), in accordance with certain embodiments of the present disclosure.

[0013] FIG. **1**D is a schematic representation of example electrical connections for pairs of electrodes embedded in an electrostatic chuck (ESC), in accordance with certain embodiments of the present disclosure.

[0014] FIG. **1**E illustrates example arrangements of capacitance sensors embedded in an electrostatic chuck (ESC), in accordance with certain embodiments of the present disclosure.

[0015] FIG. **1**F is a schematic representation of example electrical connections for pairs of electrodes embedded in an electrostatic chuck (ESC), in accordance with certain embodiments of the present disclosure.

[0016] FIG. **2**A illustrates a representation of substrate warpage, in accordance with certain embodiments of the present disclosure.

[0017] FIG. **2**B illustrates a representation of partially compensated substrate warpage, in accordance with certain embodiments of the present disclosure.

[0018] FIG. **2**C illustrates a representation of fully compensated substrate warpage, in accordance with certain embodiments of the present disclosure.

[0019] FIG. **3**A illustrates a representation of chucking/dechucking a convex portion of a substrate, in accordance with certain embodiments of the present disclosure.

[0020] FIG. **3**B illustrates a representation of chucking/dechucking a concave portion of a substrate, in accordance with certain embodiments of the present disclosure.

[0021] FIG. **3**C illustrates a representation of chucking/dechucking a concave and convex portion of a substrate, in accordance with certain embodiments of the present disclosure.

[0022] FIG. **4**A schematically illustrates a portion of a circuitry of an electrostatic chuck (ESC) with embedded electrodes, in accordance with certain embodiments of the present disclosure.

[0023] FIG. **4**B is a top view of an electrostatic chuck (ESC) with embedded electrodes, in accordance with certain embodiments of the present disclosure.

[0024] FIG. **5** is a flow diagram illustrating a method for substrate processing by chucking a substrate to a surface of a substrate support that is disposed in a processing volume, in accordance with certain embodiments of the present disclosure.

[0025] FIG. **6** illustrates an example arrangement of pairs of electrodes embedded in an electrostatic chuck (ESC), in accordance with certain embodiments of the present disclosure.

[0026] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.

DETAILED DESCRIPTION

[0027] Embodiments of the present disclosure generally relate to apparatus and methods for substrate processing and/or device packaging applications. More specifically, embodiments described herein provide for substrate and/or device packaging warpage compensation during processing. In some embodiments, a substrate or device package is chucked to a surface of a substrate support by applying a first DC bias to a first pair of electrodes and a second pair of electrodes embedded in the substrate support. For example, the first DC bias generates a first electrostatic force configured to chuck the substrate or device package to the surface of the substrate support. Embodiments of the disclosure provided herein can be useful for semiconductor manufacturing and device packaging manufacturing applications. In one semiconductor manufacturing example, a multilayer stack of thin film layers, such as a three-dimensional (3D) NAND structure formed on a substrate, can cause the substrate to bow or warp due to an intrinsic and/or an extrinsic stress formed in the layers within the multilayer stack of thin film layers. In a device packaging application example, the device package can include a three-dimensional (3D) chip stacking application in which a formed interconnect structure can cause the device package to bow or warp.

[0028] In one or more embodiments, an AC bias is additionally applied to the first pair of electrodes and the second pair of electrodes. In some examples, the AC bias causes a first AC current to flow between the first pair of electrodes and a second AC current to flow between the second pair of electrodes. If there is a difference in the first AC current and the second AC current, then there is a difference in a first capacitance formed between the first pair of electrodes and the substrate and a second capacitance formed between the second pair of electrodes and the substrate. In various embodiments, the difference in the first capacitance and the second capacitance indicates that the substrate has warped/bowed and there is a non-uniform separation between the surface of the substrate support and a portion of the substrate.

[0029] In certain embodiments, a second DC bias is applied to the first pair of electrodes and a third DC bias is applied to the second pair of electrodes. For example, the second DC bias is different from the third DC bias, and the second DC bias and the third DC bias are selected based on the difference in the first capacitance and the second capacitance. The second DC bias generates a second electrostatic force between the substrate support and a first portion of the substrate, and the third DC bias generates a third electrostatic force between the substrate support and a second portion of the substrate. In various embodiments, the second and third electrostatic forces reduce the non-uniform separation between the surface of the substrate support and the portion of the substrate. By reducing the non-uniform separation, the substrate can be processed with improved quality, reliability, and yield.

Processing System Examples

[0030] FIG. **1**A is a schematic representation of an example substrate processing chamber **100**. The substrate processing chamber **100** is representative of a variety of different systems including, without limitation, chemical vapor deposition (CVD) chambers, plasma vapor deposition (PVD) chambers, atomic layer deposition (ALD) chambers, etching chambers (including plasma-assisted systems and non-plasma-assisted systems), electron beam processing chambers, preclean chambers, thermal processing chambers, scanning electron microscope (SEM), or other similar processing systems or chambers. The substrate processing chamber **100** is illustrated to include a processing chamber **104** which contains a processing volume **106**.

[0031] As shown in FIG. **1**A, a substrate support **108** of an electrostatic chuck (ESC) is disposed within the processing volume **106**. The substrate support **108** includes a substrate supporting surface **110**, and the ESC is configured to generate one or more electrostatic forces for chucking a substrate **112** to the substrate supporting surface **110**. In some embodiments, a printed circuit board (PCB) **114** is disposed below the substrate support **108**. In other embodiments, the PCB **114** may be disposed in different orientations relative to the substrate support **108**.

[0032] In the illustrated example, the substrate support **108** includes multiple pairs of electrodes such as a first pair of electrodes **116** and a second pair of electrodes **118**. Each electrode included in the multiple pairs of electrodes is electrically coupled to a circuit layer **120** of the PCB **114**. A DC voltage source **122** and an AC voltage source **124** are also illustrated to be electrically coupled to the circuit layer **120** of the PCB **114**. In some embodiments, the DC voltage source **122** is capable of outputting example voltages of +/−750 V, +/−1500 V, +/−3000 V, etc.

[0033] The substrate processing chamber **100** is illustrated to include a controller **126** which is communicatively coupled (e.g., electrically coupled) to the circuit layer **120** of the PCB **114**. In some embodiments, the controller **126** includes a computing device having one or more processors, memory, and storage. The one or more processors can include central processing units, graphics processing units, accelerators, etc. The memory includes main memory for storing instructions for the one or more processors to execute or data for the one or more processors to operate on. For example, the memory includes random access memory (RAM). The storage includes mass storage for data or instructions. As an example and not by way of limitation, the storage may include a removable disk drive, flash memory, an optical disc, a magneto-optical disc, magnetic tape, or a Universal Serial Bus drive or two or more of these. The storage may include removable or fixed media and may be internal or external to the computing device. The storage may include any suitable form of non-volatile, solid-state memory, or read-only memory. The controller **126** includes a non-transitory computer readable medium or media. The non-transitory computer readable medium or media may include one or more semiconductor-based or other integrated circuits (ICs) (such, as for example, field-programmable gate arrays or application-specific ICs), hard disk drives, hybrid hard drives, optical discs, optical disc drives, magneto-optical discs, magneto-optical drives, solid-state drives, RAM drives, any other suitable non-transitory computer readable storage medium/media, or any suitable combination. The non-transitory computer readable medium or media may be volatile, non-volatile, or a combination of volatile and non-volatile.

[0034] The PCB **114** (e.g., the circuit layer **120**) includes multiple transistors (e.g., MOSFETs) configured as switches. In some embodiments, the controller **126** is capable of controlling the transistors included in the PCB **114** to open or close electrical connections between the DC voltage source **122** and the AC voltage source **124** and the multiple pairs of electrodes embedded in the substrate support **108**. For example, the one or more processors of the controller **126** can execute instructions which cause the one or more processors to control certain transistors included in the circuit layer **120** to close electrical connections between the DC voltage source **122** and the first pair of electrodes **116** in order to apply a first DC bias to the first pair of electrodes **116**. Similarly, executing instructions may cause the one or more processors of the controller **126** to close electrical connections between the DC voltage source **122** and the second pair of electrodes **118** in order to apply the first DC bias to the second pair of electrodes **118**. In some examples, the first DC bias can generate a first electrostatic force configured to chuck the substrate **112** to the substrate supporting surface **110**, e.g., in order to process the substrate **112**. It should be appreciated that the controller **126** is also capable of opening the electrical connections between the first pair of electrodes **116** and/or the second pair of electrodes **118** and the DC voltage source **122** to halt application of the first DC bias to the first pair of electrodes **116** and/or the second pair of electrodes **118**.

[0035] In some embodiments, the one or more processors of the controller **126** execute instructions that cause the one or more processors to control certain transistors included in the circuit layer **120** to close electrical connections between the AC voltage source **124** and the first pair of electrodes **116** in order to apply an AC bias to the first pair of electrodes **116**. Similarly, executing instructions may cause the one or more processors of the controller **126** to close electrical connections between the AC voltage source **124** and the second pair of electrodes **118** in order to apply the AC bias to the second pair of electrodes **118**. It is to be appreciated that the controller **126** is also capable of opening the electrical connections between the first pair of electrodes **116** and/or the second pair of electrodes **118** and the AC voltage source **124** to halt application of the AC bias to the first pair of electrodes **116** and/or the second pair of electrodes **118**.

[0036] The substrate processing chamber **100** is illustrated to include a vacuum source **128** in communication with the processing volume **106**. In some embodiments, a vacuum pressure from the vacuum source **128** may be utilized to flatten the substrate **112** on the substrate supporting surface **110**, e.g., before chucking the substrate **112** to the substrate supporting surface **110** via the first electrostatic force generated by the first DC bias. As shown in FIG. **1**A, the substrate processing chamber **100** includes a gas delivery system **130**, and the gas delivery system **130** is coupled to the processing volume **106**. The gas delivery system **130** is configured to deliver at least one processing gas (e.g., argon, nitrogen, oxygen, hydrogen, etc.) to the processing volume **106**. In examples in which the substrate processing chamber **100** includes a plasma-assisted system, the processing gas can include at least one of an inert gas (e.g., helium, argon, nitrogen (N.sub.2)) or dry etching gas (e.g., HBr, HF, HCl, CF.sub.4, NF.sub.3 or XeF.sub.2). In some embodiments, the gas delivery system **130** can include components for activating or energizing one or more processing gasses before delivering the processing gasses to the processing volume **106**.

[0037] In some embodiments, the substrate support **108** includes one or more resistive heating elements **180** that are coupled to a heater power supply (not shown) that is controlled by the controller **126**. For simplicity of discussion, heater power supply and resistive heating elements **180** are referred to herein as a heater assembly **181**. As shown, the one or more heating elements **180** may be disposed below and be electrically isolated from each of the first pair of electrodes **116** and the second pair of electrodes **118** within the substrate support **108**. In various embodiments, the heater assembly **181** facilitates the temperature control of the substrate support **108** for thermal stability and thermal stress management of a substrate disposed on the substrate supporting surface **110**.

[0038] FIG. **1**B illustrates an example **102** of substrate warpage occurring during substrate processing. Ensuring the flatness of the substrate **112** on the substrate supporting surface **110** during the substrate processing improves quality, reliability, and yield of manufactured devices that are formed on the substrate **112**. However, maintaining the flatness of the substrate **112** on the substrate supporting surface **110** of the substrate support **108** is challenging because temperatures within the processing volume **106** can range from −90 degrees Celsius to more than 500 degrees Celsius. These variations in temperature during the substrate processing affect the stresses formed in substrate causing the substrate **112** to warp or bow. As shown in FIG. **1**B, the bowed substrate **112** is no longer flat on the substrate supporting surface **110** of the substrate support **108**. Instead, there is a non-uniform separation **132** between the substrate supporting surface **110** and portions of the substrate **112** which would result in a diminished quality/yield of the manufactured devices that are formed on the substrate **112**.

[0039] In some embodiments, in order to reduce the non-uniform separation **132** between the substrate supporting surface **110** and the substrate **112** by at least a threshold amount (e.g., 50 percent, 60 percent, 70 percent, 80 percent, between 30 and 80 percent, etc.), the one or more processors of the controller **126** execute instructions which cause the one or more processors to apply the AC bias to the first pair of electrodes **116** and the second pair of electrodes **118**. For example, the AC bias causes AC current to flow between the first pair of electrodes **116** and between the second pair of electrodes **118**. Since the substrate **112** is a first distance (e.g., a first average distance) from a portion of the substrate supporting surface **110** above the first pair of electrodes **116** and the substrate **112** is a second distance (e.g., a second average distance) from a portion of the substrate supporting surface **110** above the second pair of electrodes **118**, a first capacitance forms between the first pair of electrodes **116** and the substrate **112** and a second capacitance forms between the second pair of electrodes **116** and the substrate **112**. For this reason, a first AC current flows between the first pair of electrodes **116** and a second AC current flows between the second pair of electrodes **118**.

[0040] In some embodiments, a first electric flow is electricity flowed between the first pair of electrodes **116** and a second electric flow is electricity flowed between the second pair of electrodes **118**. The first electric flow includes electrical characteristics. The second electric flow also includes electrical characteristics. Examples of electrical characteristics may include one or more of capacitance, voltage, current, AC current, DC current, frequency, inductance, and amplitude. For example, the controller **126** is able to detect changes in capacitance by measuring and/or detecting electrical characteristics of the first electric flow between the first pair of electrodes **116**. Further the controller **126** is able to detect changes in current by measuring and/or detecting electrical characteristics of the first electric flow between the first pair of electrodes **116**.

[0041] In some embodiments, the one or more processors of the controller **126** execute instructions that cause the one or more processors to detect a difference between the first AC current and the second AC current and/or to detect a difference between the first capacitance and the second capacitance. For example, by use of the methods described herein to reduce the substrate bow will reduce the difference between the first capacitance and the second capacitance due to the reduction in the non-uniform separation **132**. Instructions executed by the one or more processors of the controller **126** cause the one or more processors to apply a second DC bias to the first pair of electrodes **116** and to apply a third DC bias to the second pair of electrodes **118**. In some embodiments, the second DC bias is different from the third DC bias, and the second DC bias and the third DC bias are selected based on detecting the difference between the first capacitance and the second capacitance. In one or more embodiments, the second DC bias generates a second electrostatic force between the substrate support **108** and the substrate **112** and the third DC bias generates a third electrostatic force between the substrate support **108** and the substrate **112**. For example, the second electrostatic force may have a greater magnitude than the third electrostatic force because the first distance (e.g., the first average distance) of the substrate **112** from the portion of the substrate supporting surface **110** above the first pair of electrodes **116** is greater than the second distance (e.g., the second average distance) of the substrate **112** from the portion of the substrate supporting surface **110** above the second pair of electrodes **118**.

[0042] FIG. **1**C illustrates example arrangements of pairs of electrodes embedded in an electrostatic chuck (ESC). As shown, FIG. **1**C illustrates views of the substrate supporting surface **110** from above the substrate supporting surface **110** which depict example arrangements of pairs of electrodes **134** embedded in the substrate support **108**. For example, substrate supporting surface

110-1 depicts the pairs of electrodes **134** arranged in linear rectangular array. In some embodiments, the pairs of electrodes **134** may be arranged non-linearly. In the illustrated example, substrate supporting surface **110-2** includes the pairs of electrodes **134** arranged in a concentric array. Substrate supporting surface **110-3** depicts the pairs of electrodes **134** that are arranged in a linear rectangular array that are further arranged or grouped in regions **136**, **138**, **140**, and **142**. In the illustrated example, substrate supporting surface **110-4** includes an inner pair of electrodes **137***a* and outer electrodes **137***b* arranged around the inner pair of electrodes **137***a*. It is to be appreciated that the pairs of electrodes **134** may be disposed within the substrate support **108** in a variety of different arrangements and that the surfaces **110-1**, **110-2**, and **110-3** illustrated in FIG. **1**C are intended to be non-limiting examples.
[0043] FIG. **1**D is a schematic representation of example electrical connections for pairs of electrodes embedded in an electrostatic chuck (ESC). In the illustrated example, the substrate support **108** includes pairs of electrodes **134-1** to **134-7**. Although seven pairs of electrodes **134** are illustrated, it is to be appreciated that, in various embodiments, the substrate support **108** may include 50 pairs of electrodes **134**, 100 pairs of electrodes **134**, 150 pairs of electrodes **134**, 200 pairs of electrodes **134**, 250 pairs of electrodes **134**, etc. Each of the pairs of electrodes **134-1** to **134-7** includes a positive electrode and a negative electrode.
[0044] In some embodiments, instead of including two electrical connections to the circuit layer **120** of the PCB **114** for each of the pairs of electrodes **134-1** to **134-7**, the substrate support **108** may include a common electrical connection **144**. In one or more examples, the common electrical connection **144** connects the negative electrode (or the positive electrode) included in each of the pairs of electrodes **134-1** to **134-7** to the circuit layer **120**. As shown in FIG. **1**D, an electrical connection **146** connects the positive electrode included in the pair of electrodes **134-1** to the circuit layer **120** of the PCB **114**. As further shown, an electrical connection **148** connects the negative electrode included in the pair of electrodes **134-1** to the common electrical connection **144** which is connected to the circuit layer **120**. In some embodiments, the common electrical connection **144** is couple to a ground reference that is coupled to a circuit formed in the PCB **114**. In some embodiments, two pairs of electrodes of the pairs of electrodes **134-1-134-7** are used to detect capacitance differences and the rest of the pairs of electrodes detect differences in current. For example, the pair of electrodes **134-1** detects a first capacitance, the pair of electrodes **134-2** detects a second capacitance, and the pairs of electrodes **134-3** to **134-7** are used to chuck the substrate **112**. In another embodiment that make be combined with other embodiments, the pair of electrodes **134-1** detects capacitance and with 1 pair and pair of electrodes **134-2** detects high voltage such that 1 pair of electrodes determines is the substrate is in contact with an electrode a the other electrode pair can be used to detect the distance the substrate is deflected from the substrate supporting surface **110**.
[0045] In certain embodiments, an electrical connection **150** connects the positive electrode included in the pair of electrodes **134-2** to the circuit layer **120**, and an electrical connection **152** connects the negative electrode included in the pair of the electrodes **134-2** to the common electrical connection **144**. For instance, the positive electrode and the negative electrode included in the pair of electrodes **134-3** are connected by electrical connections **154**, **156** to the circuit layer **120** and the common electrical connection **144**, respectively. In another example, the positive electrode included in the pair of electrodes **134-4** is connected to the circuit layer **120** of the PCB **114** by an electrical connection **158**, and the negative electrode included in the pair of electrodes **134-4** is connected to the common electrical connection **144** by an electrical connection **160**.
[0046] Similarly, the positive electrodes included in pairs of electrodes **134-5**, **134-6**, **134-7** are connected to the circuit layer **120** by electrical connections **162**, **166**, **170**, respectively. The negative electrodes included in the pairs of electrodes **134-5**, **134-6**, **134-7** are connected to the common electrical connection **144** by electrical connections **164**, **168**, **172**, respectively. In some embodiments, the circuit layer **120** includes at least one transistor (e.g., a MOSFET) configured as

a switch for each of the pairs of electrodes **134-1-134-7** and for the common electrical connection **144** so as to selectively connect and disconnect each of the electrodes to a power sources (e.g., DC voltage source **122** or AC voltage source **124**) or ground reference as desired. In some embodiments, one terminal of a switch is coupled to a positive electrode that is coupled to a positive electrode connection (e.g., one of the electrical connections **146**, **150**, **154**, **158**, **162**, **166** and **170**) and the second terminal of the switch is coupled to one or more of the power sources, such as DC voltage source **122** and AC voltage source **124**. Similarly, in this configuration, one terminal of a switch is coupled to the negative electrodes that are each coupled to a negative electrode connection (e.g., one of the electrical connections **148**, **152**, **156**, **160**, **164**, **168** and **172**) and the second terminal of the switch is coupled to a ground reference. In one or more embodiments, the one or more processors of the controller **126** execute instructions which cause the one or more processors to operate the transistors configured as switches included in the circuit layer **120** of the PCB **114** such that any DC bias or AC bias can be applied between any of the pairs of electrodes **134-1-134-7**. In various embodiments, leveraging the common electrical connection **144** and the transistors configured as switches, an additional electrical connection to the circuit layer **120** can be avoided for each of the pairs of electrodes **134-1** to **134-7**.

[0047] FIG. **1**E illustrates example arrangements of capacitance sensors embedded in an electrostatic chuck (ESC). In some embodiments, the substrate support **108** includes capacitance sensors **174** disposed below the substrate supporting surface **110**. The capacitance sensors **174** are illustrated to be electrically connected to a portion of the circuit layer **120** of the PCB **114** that is coupled to controller **126**. In one or more embodiments, the controller **126** receives values measured by the capacitance sensors **174** which correspond to distances between portions of the substrate **112** and the substrate supporting surface **110**. In various embodiments, each of the capacitance sensors **174** is associated with a unique identifier which is also associated with a particular location on the substrate supporting surface **110**.

[0048] In some embodiments, the capacitance sensors **174** are capable of indicating a distance between a portion of the substrate **112** and a portion of the surface **110** corresponding to one of the pairs of electrodes **134-1-134-7**. In one or more embodiments, the controller **126** utilizes the distances corresponding to each of the pairs of electrodes **134-1-134-7** to sequentially apply DC biases to the pairs of electrodes **134-1** to **134-7**. In certain embodiments, the controller **126** applies a relatively high DC bias to whichever one of the pairs of electrodes **134-1-134-7** is associated with a smallest (e.g., closest) distance. In various embodiments, the controller **126** then reduces the relatively high DC bias to a holding bias, and applies the relatively high DC bias to whichever one of the pairs of electrodes **134-1-134-7** is associated with a next smallest (e.g., next closest) distance. In one or more embodiments, the controller **126** then reduces the relatively high DC bias to the holding bias and repeats this process until the substrate **112** is flat on the surface **110**.

[0049] In some embodiments, the controller **126** applies DC biases to the pairs of electrodes **134-1-134-7** in a sequential manner, e.g. a sequential order, based on a bow of the substrate **112** (e.g., convex or concave). In certain embodiments, the controller **126** applies DC biases to the pairs of electrodes **134-1-134-7** base on a portion of the substrate **112** that initially contacts the surface **110**. In various embodiments, the controller **126** applies DC biases to the pairs of electrodes **134-1** to **134-7** in a manner configured to minimize an amount of movement of the substrate **112** relative to the substrate support **108** (e.g., during dechucking).

[0050] FIG. **1**F is a schematic representation of example electrical connections for electrodes embedded in an electrostatic chuck (ESC). In the illustrated example, the substrate support **108** includes electrodes **135-1** to **135-7**. Although seven electrodes **135** are illustrated, it is to be appreciated that, in various embodiments, the substrate support **108** may include 50 electrodes **135**, 100 electrodes **135**, 150 electrodes **135**, 200 electrodes **135**, **250** electrodes **135**, etc. Each of the electrodes **135-1** to **135-7** includes a positive electrode or a negative electrode. The electrodes **135-1** to **135-7** pair with a common electrode **137**. While shown as electrodes **135-1** to **135-7** being positive and the common electrode **137** being negative, the electrodes **135-1** to **135-7** could be negative and the common electrode **137** being positive.

[0051] Each electrode **135** and the common electrode is connected to the circuit layer **120** of the PCB **114**. As shown in FIG. **1**F, an electrical connection **146** connects the electrode **135-1** to the circuit layer **120** of the PCB **114**. In some embodiments, the electrical connection **139** is coupled to a ground reference that is coupled to a circuit formed in the PCB **114**

[0052] In certain embodiments, the electrical connection **150** connects the electrode **135-2** to the circuit layer **120**, and an electrical connection **139** connects the common electrode **137** to the circuit layer **120**. For instance, the electrode **135-3** is connected by electrical connection **154** to the circuit layer **120**. In another example, the electrode **135-4** is connected to the circuit layer **120** of the PCB **114** by an electrical connection **158**, and similarly, the electrodes **135-5**, **135-6**, and **135-7** are connected to the circuit layer **120** by electrical connections **162**, **166**, **170**, respectively.

[0053] In some embodiments, the circuit layer **120** includes at least one transistor (e.g., a MOSFET) configured as a switch for each of the electrodes **135-1-135-7** and for the electrical connection **139** so as to selectively connect and disconnect each of the electrodes to a power sources (e.g., DC voltage source **122** or AC voltage source **124**) or ground reference as desired. In some embodiments, one terminal of a switch is coupled to a positive electrode that is coupled to a positive electrode connection (e.g., one of the electrical connections **146**, **150**, **154**, **158**, **162**, **166** and **170**) and the second terminal of the switch is coupled to one or more of the power sources, such as DC voltage source **122** and AC voltage source **124**. Similarly, in this configuration, one terminal of a switch is coupled to the negative electrodes that are each coupled to the common electrode connection **139** and the second terminal of the switch is coupled to a ground reference. In one or more embodiments, the one or more processors of the controller **126** execute instructions which cause the one or more processors to operate the transistors configured as switches included in the circuit layer **120** of the PCB **114** such that any DC bias or AC bias can be applied between any of the electrodes **135-1** to **135-7** and the common electrode **137**. In various embodiments, leveraging the electrical connection **139** and the transistors configured as switches, an additional electrical connection to the circuit layer **120** can be avoided for each of the electrodes **135-1** to **135-7**.

Substrate Warpage Compensation Examples

[0054] FIG. **2**A illustrates a representation **200** of substrate warpage. As shown, the substrate **112** has warped (e.g., due to temperature variations within the processing volume **106**), and the substrate **112** is no longer flat relative to the substrate supporting surface **110** of the electrostatic chuck (ESC). In some embodiments, one or more forces warping the substrate **112** are greater than the first electrostatic force generated by applying the first DC bias to the first pair of electrodes **116** and the second pair of electrodes **118**. For example, a portion of the substrate **112** is bowed such that the portion of the substrate **112** is separated from the substrate supporting surface **110** of the substrate support **108** by the non-uniform separation **132**.

[0055] The first pair of electrodes **116** is illustrated to include a first electrode **206** and a second electrode **208**, and the second pair of electrodes **118** is illustrated to include a third electrode **210** and a fourth electrode **212**. As shown in FIG. **2**A, the first electrode **206** and the second electrode **208** are each associated with a first conductor **214**, a second conductor **216**, and a third conductor **218**. Similarly, the third electrode **210** and the fourth electrode **212** are also each associated with a first conductor **214**, a second conductor **216**, and a third conductor **218**. In some examples, the one or more processors of the controller **126** execute instructions which cause the one or more processors to supply a DC voltage from the DC voltage source **122** via the first conductors **214**, supply an AC voltage from the AC voltage source **124** via the second conductors **216**, and measure an AC current and/or capacitance via the third conductors **218**.

[0056] In various embodiments, the non-uniform separation **132** between a portion of the substrate and the substrate supporting surface **110** is detectable by the capacitance sensors **174**. In some embodiments, in order to detect the non-uniform separation **132**, the one or more processors of the controller **126** execute instructions that cause the one or more processors to apply an AC bias (e.g., via the second conductors **216**) to the first pair of electrodes **116** and to the second pair of electrodes **118**. For example, applying the AC bias to the first pair of electrodes **116** causes a first AC current I.sub.AC1 to flow between the first electrode **206** and the second electrode **208**. Similarly, applying the AC bias to the second pair of electrodes **118** causes a second AC current I.sub.AC2 to flow between the third electrode **210** and the fourth electrode **212**. In various examples, the one or more processors of the controller **126** execute instructions that cause the one or more processors to measure the first AC current I.sub.AC1 and the second AC current I.sub.AC2 (e.g., via the third conductors **218**).

[0057] If the first AC current I.sub.AC1 is different from the second AC current I.sub.AC2 based on application of the same AC bias to the first and second pairs of electrodes **116**, **118**, then a first capacitance **220** is different from a second capacitance **222**. In some embodiments, the first capacitance **220** corresponds to a first distance (e.g., a first average distance) between a portion of the substrate supporting surface **110** above the first pair of electrodes **116** and a first portion of the substrate **112**. In these embodiments, the second capacitance **222** corresponds to a second distance (e.g., a second average distance) between a portion of the substrate supporting surface **110** above the second pair of electrodes **118** and a second portion of the substrate **112**. Accordingly, a difference between the first capacitance **220** and the second capacitance **222** indicates that the first distance (e.g., the first average distance) is different from the second distance (e.g., the second average distance). Therefore, the substrate **112** is not flat on the substrate supporting surface **110** of the substrate support **108**. For similar reasons, a difference between the first AC current I.sub.AC1 and the second AC current I.sub.AC2 also indicates that the first distance is different from the second distance and that the substrate **112** is not flat on the substrate supporting surface **110**.

[0058] In certain embodiments, the one or more processors of the controller **126** execute instructions which cause the one or more processors to apply a second DC bias (e.g., via the first conductors **214**) to the first pair of electrodes **116** and to apply a third DC bias (e.g., via the first conductors **214**) to the second pair of electrodes **118**. In some embodiments, the second DC bias may be in a range of 500 V to 4000 V and the third DC bias may be in a range of 500 V to 4000 V. For example, the second DC bias generates a second electrostatic force between the portion of the substrate supporting surface **110** above the first pair of electrodes **116** and the first portion of the substrate **112**. In this example, the second electrostatic force reduces the first distance between the portion of the substrate supporting surface **110** above the first pair of electrodes **116** and the first portion of the substrate **112**. In some examples, the third DC bias generates a third electrostatic force between the portion of the substrate supporting surface **110** above the second pair of electrodes **118** and the second portion of the substrate **112**. In these examples, the third electrostatic force reduces the second distance between the portion of the substrate supporting surface **110** above the second pair of electrodes **116** and the second portion of the substrate **112**.

[0059] In some embodiments of the substrate support **108**, a vacuum pressure from the vacuum source **128**, which is in communication with the processing volume **106** and/or an array of vacuum ports **155** (FIG. **1**C) that are coupled to the substrate supporting surface **110** may be utilized to reduce the first distance between the portion of the substrate supporting surface **110** above the first pair of electrodes **116** and the first portion of the substrate **112** and/or to reduce the second distance between the portion of the substrate supporting surface **110** above the second pair of electrodes **116** and the second portion of the substrate **112**. In one or more embodiments, the vacuum pressure generated by the vacuum source **128** may be used to create a pressure differential that is used to reduce the first and second distances in addition or alternative to the second and third electrostatic forces. Similarly, the vacuum pressure from the vacuum source **128** can be used to reduce the difference between the first AC current lac and the second AC current I.sub.AC2 and/or to reduce the difference between the first capacitance **220** and the second capacitance **222**. In some embodiments, the vacuum pressure from the vacuum source **128** and/or the first and second electrostatic forces may reduce the difference between AC current I.sub.AC1 and the second AC current I.sub.AC2 between 5 percent and 80 percent and may reduce the difference between the first capacitance **220** and the second capacitance **222** between 5 percent and 50 percent. Although examples are described with respect to the vacuum pressure from the vacuum source **128** and the second and third electrostatic forces, it is to be appreciated that other types of systems/forces can be used to reduce the first and second distances in various embodiments. For example, the substrate processing chamber **100** can include additional vacuum sources (e.g., vacuum pressures), one or more actuators (e.g., applied forces), one or more clamps (e.g., compressive forces), and/or one or more mechanical springs (e.g., spring forces).

[0060] FIG. 2B illustrates a representation **202** of partially compensated substrate warpage. As shown the representation **202**, the second electrostatic force and the third electrostatic force have reduced the non-uniform separation **132** relative to the non-uniform separation **132** of the representation **200**. However, the substrate **112** is still not flat relative to the substrate supporting surface **110** of the ESC.

[0061] A third capacitance **224** corresponds to a third distance (e.g., a third average distance) between the portion of the substrate supporting surface **110** above the first pair of electrodes **116** and the first portion of the substrate **112**. A fourth capacitance **226** corresponds to a fourth distance (e.g., a fourth average distance) between the portion of the substrate supporting surface **110** above the second pair of electrodes **118** and the second portion of the substrate **112**. In some embodiments, a difference between the third capacitance **224** and the fourth capacitance **226** indicates that the third distance (e.g., the third average distance) is different from the fourth distance (e.g., the fourth average distance). As a result, the one or more processors of the controller **126** can execute instructions which cause the one or more processors to determine that the substrate **112** is not flat on the substrate supporting surface **110**. Similarly, a difference between the first AC current I.sub.AC1 and the second AC current I.sub.AC2 also indicates that the third distance is different from the fourth distance and that the substrate **112** is not flat on the substrate supporting surface **110**. In various embodiments, the one or more processors of the controller **126** execute instructions that cause the one or more processors to detect the difference in the third capacitance **224** and the fourth capacitance **226** by measuring the difference between the first AC current I.sub.AC1 and the second AC current I.sub.AC2 and using that difference to make a determination as to whether the substrate **112** is or is not flat on the substrate supporting surface **110**.

[0062] In various embodiments, the one or more processors of the controller **126** execute instructions which cause the one or more processors to apply a fourth DC bias to the first pair of electrodes **116** and to apply a fifth DC bias to the second pair of electrodes **118**. In some embodiments, the fourth DC bias may be in a range of 100 V to 5000 V and the fifth DC bias may be in a range of 100 V to 5000 V. In certain embodiments, instructions executed by the one or more processors of the controller **126** cause the one or more processors to detect a change in the difference in the first capacitance **220** and the second capacitance **222** based on the difference in the third capacitance **224** and the fourth capacitance **226**. In these embodiments, the one or more processors of the controller **126** apply the forth DC bias to the first pair of electrodes **116** and the fifth DC bias to the second pair of electrodes **118** in response to detecting the change in the difference in the first capacitance **220** and the second capacitance **222**.

[0063] The fourth DC bias generates a fourth electrostatic force between the portion of the substrate supporting surface **110** above the first pair of electrodes **116** and the first portion of the substrate **112**. The fourth electrostatic force reduces the third distance between the portion of the substrate supporting surface **110** above the first pair of electrodes **116** and the first portion of the substrate **112**. For example, the fifth DC bias generates a fifth electrostatic force between the portion of the substrate supporting surface **110** above the second pair of electrodes **118** and the second portion of the substrate **112**. In this example, the fifth electrostatic force reduces the fourth distance between the portion of the substrate supporting surface **110** above the second pair of electrodes **116** and the second portion of the substrate **112**.

[0064] FIG. **2**C illustrates a representation **204** of fully compensated substrate warpage. As shown, the fourth electrostatic force and the fifth electrostatic force have reduced the non-uniform separation **132** such that the substrate **112** is flat relative to the substrate supporting surface **110** of the substrate support **108**. In some embodiments, the first pair of electrodes **116** is associated with a capacitance **228** and the second pair of electrodes **118** is also associated with the capacitance **228**. For example, the one or more processors of the controller **126** execute instructions that cause the one or more processors to compute a difference between the capacitance **228** and the capacitance **228** which is minimal and indicates that the substrate **112** is flat relative to the substrate supporting surface **110** of the ESC. In some examples, the one or more processors of the controller **126** determine that the difference between the first capacitance **220** and the second capacitance **222** has been reduced by at least a threshold amount (e.g., 70 percent, 80 percent, 90 percent, between 5 percent and 50 percent, etc.).

[0065] In one or more embodiments, the one or more processors of the controller **126** may execute instructions that cause the one or more processors to halt application of the fourth DC bias to the first pair of electrodes **116** and to halt application of the fifth DC bias to the second pair of electrodes **118**. For example, the first electrostatic force generated by the first DC bias may chuck the substrate **112** to the substrate supporting surface **110** of the substrate support **108** in order to process the substrate **112**. By reducing the non-uniform separation **132** the quality, reliability, and yield of manufactured devices that include the substrate **112** are improved.

Substrate Chucking/Dechucking Examples

[0066] FIG. **3**A illustrates a representation **300** of chucking/dechucking a convex portion of a substrate. In order to chuck a convex substrate **306**, the one or more processors of the controller **126** execute instructions that cause the one or more processors to apply a center (e.g., the portion of the convex substrate **306** contacting the substrate supporting surface **110**) to edge (e.g., the portion of the convex substrate **306** separated from the substrate supporting surface **110** by the greatest distance) voltage using an increase decrease algorithm. In some examples, the second pair of electrodes **118** applies a relatively low voltage to hold the convex substrate **306** while the first pair of electrodes **116** applies a relatively high voltage to flatten the convex substrate **306**. In order to dechuck the convex substrate **306**, a reverse voltage waveform is used to dechuck the convex substrate **306** controllably.

[0067] FIG. **3**B illustrates a representation **302** of chucking/dechucking a concave portion of a substrate. In order to chuck a concave substrate **308**, the one or more processors of the controller **126** execute instructions that cause the one or more processors to apply a relatively high voltage at an edge of the substrate supporting surface **110**, and then to sweep relatively high voltages and relatively low voltages to roll the concave substrate **308** onto the chuck from the edge of the substrate supporting surface **110** to the other edge of the substrate supporting surface **110**. In various embodiments, in order to dechuck the concave substrate **308**, a reverse voltage waveform is used to dechuck the concave substrate **308** controllably.

[0068] FIG. **3**C illustrates a representation **304** of chucking/dechucking a concave and convex portion of a substrate. In order to chuck a concave and convex substrate **310**, the one or more processors of the controller **126** execute instructions that cause the one or more processors to start applying voltages to convex points of the concave and convex substrate **310** and then proliferate the applied voltages. For example, the concave and convex substrate **310** can include a “wave” shape as illustrated or a “saddle” shape which is convex between first and second ends and concave between third and fourth ends. In one or more embodiments, in order to dechuck the concave and convex substrate **310**, a reverse voltage waveform is used to dechuck the concave and convex substrate **310** controllably.

[0069] FIGS. **4**A and **4**B schematically illustrate portions of an electrostatic chuck (ESC) that includes embedded bipolar electrodes. FIG. **4**A illustrates a representation **400** of the electrode circuitry that is in communications with the controlling circuits in the circuit layer **120** of the PCB **114** disposed within a substrate support. FIG. **4**B illustrates a top view **401** of the surface **402** of the substrate support. The surface **402** of the substrate support **108** is partially covered with a conductive material (e.g., metal) over a majority of an area of the surface **402**. In various embodiments, the conductive material covers 95 percent of the surface **402**, 90 percent of the surface **402**, 85 percent of the surface **402**, etc. In certain embodiments, a plurality of regions, such as regions **436**, **438**, **440**, **442**, **444**, **446**, **448**, **450**, are not covered by the conductive material. In some embodiments, the surface **402** has a first surface area and the regions **436**, **438**, **440**, **442**, **444**, **446**, **448**, **450** collectively have a second surface area such that the second surface area is less than about 15 percent of the first surface area, less than about 10 percent of the first surface area, less than about 5 percent of the first surface area, etc.

[0070] In various embodiments, a pair of bipolar electrodes **404**, **405** and a pair of bipolar electrodes **406**, **407** are included in the region **436** such that conductors of the pair of bipolar electrodes **404**, **405** are interdigitated and conductors of the pair of bipolar electrodes **406**, **407** are interdigitated. The region **438** includes a pair of bipolar electrodes **408**, **409** with interdigitated conductors and a pair of bipolar electrodes **410**, **411** with interdigitated conductors. A pair of bipolar electrodes **412**, **413** with interdigitated conductors and a pair of bipolar electrodes **414**, **415** with interdigitated conductors are included in the region **440**. In some examples, the region **442** includes a pair of bipolar electrodes **416**, **417** with interdigitated conductors and a pair of bipolar electrodes **418**, **419** with interdigitated conductors. In one or more embodiments, the region **444** includes a pair of bipolar electrodes **420**, **421** with interdigitated conductors and a pair of bipolar electrodes **422**, **423** with interdigitated conductors. A pair of bipolar electrodes **424**, **425** with interdigitated conductors and a pair of bipolar electrodes **426**, **427** are included in the region **446**. The region **448** includes a pair of bipolar electrodes **428**, **429** with interdigitated conductors and a pair of bipolar electrodes **430**, **431** with interdigitated conductors. In various examples, a pair of bipolar electrodes **432**, **433** with interdigitated conductors and a pair of bipolar electrodes **434**, **435** with interdigitated conductors are included in the region **450**. Each of the pairs of bipolar electrodes are configured to apply an electrical bias between each of the interdigitated electrodes to electrostatically chuck a portion of a substrate disposed over the pair of interdigitated electrodes.

[0071] In one example, as shown in FIGS. **4**A-**4**B, the region **436** contains the pair of bipolar electrodes **404**, **405** and a pair of electrodes **406**, **407**; the region **438** contains the pair of bipolar electrodes **408**, **409** and a pair of electrodes **410**, **411**; the region **440** contains the pair of bipolar electrodes **412**, **413** and a pair of electrodes **414**, **415**; the region **442** contains the bipolar electrodes **416**, **417** and a pair of electrodes **418**, **419**; the region **444** contains the bipolar electrodes **420**, **421** and a pair of electrodes **422**, **423**; the region **446** contains the bipolar electrodes **424**, **425** and a pair of electrodes **426**, **427**; the region **448** contains the bipolar electrodes **428**, **429** and a pair of electrodes **430**, **431**; and the region **450** contains the bipolar electrodes **432**, **433** and a pair of electrodes **434**, **435**.

[0072] In one or more embodiments, the bipolar electrodes **405**, **409**, **413**, **417**, **421**, **425**, **429**, **433** have a first polarity (e.g., positive or negative) and the bipolar electrodes **407**, **411**, **415**, **419**, **423**, **427**, **431**, **435** also have the first polarity (e.g., positive or negative) when biased. In some embodiments, the bipolar electrodes **404**, **408**, **412**, **416**, **420**, **424**, **428**, **432** have a second polarity (e.g., positive or negative) and the bipolar electrodes **406**, **410**, **414**, **418**, **422**, **426**, **430**, **434** also have the second polarity (e.g., positive or negative) when biased. In certain embodiments, the bipolar electrodes **405**, **409**, **413**, **417**, **421**, **425**, **429**, **433** have a first polarity (e.g., positive or negative) and the bipolar electrodes **404**, **408**, **412**, **416**, **420**, **424**, **428**, **432** also have the first polarity (e.g., positive or negative) when biased. In various embodiments, the bipolar electrodes **407**, **411**, **415**, **419**, **423**, **427**, **431**, **435** also have a second polarity (e.g., positive or negative) and the bipolar electrodes **406**, **410**, **414**, **418**, **422**, **426**, **430**, **434** also have the second polarity (e.g., positive or negative) when biased.

[0073] In some embodiments, one of the bipolar electrodes contained in each of the regions **436**, **438**, **440**, **442**, **444**, **446**, **448**, **450** is part of a first chucking group and the other one of the bipolar electrodes contained in each of the regions **436**, **438**, **440**, **442**, **444**, **446**, **448**, **450** is part of a second chucking group. The first chucking group includes the pairs of bipolar electrodes **404-405**, **408-409**, **412-413**, **416-417**, **420-421**, **424-425**, **428-429**, and **432-433** and the second chucking group includes the bipolar electrodes **406-407**, **410-412**, **414-415**, **418-419**, **422-423**, **426-427**, **430-431**, and **434-435**. In certain embodiments, the bipolar electrodes included in the first chucking group are configured to be separately biasable relative to the bipolar electrodes included in the second chucking group. Similarly, in some embodiments, the bipolar electrodes included in the second chucking group are configured to be separately biasable relative to the bipolar electrodes included in the first chucking group.

[0074] In certain embodiments, it may be desirable to avoid generating an electric field within a portion of one of the regions **436**, **438**, **440**, **442**, **444**, **446**, **448**, **450** when chucking/dechucking the substrate **112** to the surface **402**. For example, in some SEM type of applications, a scanning electron beam may be directed in close proximity to a portion of one of the pairs of bipolar electrodes in one of the regions **436**, **438**, **440**, **442**, **444**, **446**, **448**, **450**. If it becomes desirable to avoid generating an electric field by a particular one of the pairs of bipolar electrodes when chucking/dechucking the substrate **112**, then the one or more processors of the controller **126** execute instructions which cause the one or more processors to chuck/dechuck the substrate **112** using whichever one of the first and second chucking groups that does not include the particular one of the pairs of bipolar electrodes. In an example in which it becomes desirable to avoid generating an electric field by the bipolar electrodes **408**,**409**, then the controller **126** uses the second chucking group which does not include the bipolar electrodes **408**,**409** to chuck/dechuck the substrate **112** and disconnects the bipolar electrodes in the first chucking group from a power source. In another example in which it becomes desirable to avoid generating an electric field by the bipolar electrodes **405**, **407**, then the controller **126** uses the first chucking group which does not include the bipolar electrodes **405**, **407** to chuck/dechuck the substrate **112**. In various embodiments, the bipolar electrodes **405**, **409**, **413**, **417**, **421**, **425**, **429**, **433**, **407**, **411**, **415**, **419**, **423**, **427**, **431**, **435** are configured to chuck/dechuck the substrate. In some embodiments, the bipolar electrodes **412**, **416**, **420**, **424**, **428**, **432**, **406**, **410**, **414**, **418**, **422**, **426**, **430**, **434** are configured to chuck/dechuck the substrate **112**.

[0075] FIG. **5** is a flow diagram illustrating a method **500** for substrate processing by chucking a substrate to a surface of a substrate support that is disposed in a processing volume. At **502**, a first DC bias is applied to a first pair of electrodes and a second pair of electrodes, the first pair of electrodes disposed in a first portion of a substrate support and the second pair of electrodes disposed in a second portion of the substrate support. In one or more embodiments, the one or more processors of the controller **126** apply the first DC bias to the first pair of electrodes **116** and the second pair of electrodes **118**.

[0076] At **504**, an AC bias is applied to the first pair of electrodes and the second pair of electrodes. In some embodiments, the one or more processors of the controller apply the AC bias to the first pair of electrodes **116** and the second pair of electrodes **118**. At **506**, a difference is detected in a first capacitance formed between the first pair of electrodes and the substrate and a second capacitance formed between the second pair of electrodes and the substrate. In various embodiments, the one or more processors of the controller **126** detect the difference in the first capacitance **220** and the second capacitance **222**.

[0077] At **508**, a second DC bias is applied to the first pair of electrodes and a third DC bias is applied to the second pair of electrodes, wherein the second DC bias is different from the third DC bias, and the second DC bias and the third DC bias are selected based on detecting the difference in the first capacitance and the second capacitance. In some embodiments, the one or more processors of the controller apply the second DC bias to the first pair of electrodes **116** and the third DC bias to the second pair of electrodes **118**. For example, the second DC bias and the third DC bias are selected based on detecting the difference in the first capacitance **220** and the second capacitance **222** in order to minimize the difference between the first capacitance **220** and the second capacitance **222**.

[0078] FIG. **6** illustrates example arrangements of pairs of electrodes embedded in an electrostatic chuck (ESC) **601**. As shown, FIG. **6** illustrates a view of the substrate supporting surface **603** of a rectangular ESC from above the substrate supporting surface **603** with an example arrangement electrodes embedded in the ESC **601**. For example, substrate supporting surface **603** depicts the pairs of electrodes arranged in a rectangular array. In some embodiments, the pairs of electrodes include inner electrode pairs **607**, lateral electrodes pairs **605**, and corner electrodes **609**. It is to be appreciated that the inner electrode pair **607**, lateral electrodes pairs **605**, and corner electrodes **609** may be disposed within the substrate supporting surface **603** in a variety of different arrangements and that the surface **603** illustrated in FIG. **6** is intended to be non-limiting examples.

Additional Considerations

[0079] In the above description, details are set forth by way of example to facilitate an understanding of the disclosed subject matter. It should be apparent to a person of ordinary skill in the field, however, that the disclosed implementations are exemplary and not exhaustive of all possible implementations. Thus, it should be understood that reference to the described examples is not intended to limit the scope of the disclosure. Any alterations and further modifications to the described devices, instruments, methods, and any further application of the principles of the present disclosure are fully contemplated as would normally occur to one skilled in the art to which the disclosure relates. In particular, it is fully contemplated that the features, components, and/or steps described with respect to one implementation may be combined with the features, components, and/or steps described with respect to other implementations of the present disclosure. As used herein, the term “about” may refer to a +/−10% variation from the nominal value. It is to be understood that such a variation can be included in any value provided herein.

[0080] As used herein, “a processor,” “at least one processor” or “one or more processors” generally refers to a single processor configured to perform one or multiple operations or multiple processors configured to collectively perform one or more operations. In the case of multiple processors, performance of the one or more operations could be divided amongst different processors, though one processor may perform multiple operations, and multiple processors could collectively perform a single operation. Similarly, “a memory,” “at least one memory” or “one or more memories” generally refers to a single memory configured to store data and/or instructions, multiple memories configured to collectively store data and/or instructions.

[0081] As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover: a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c).

[0082] The methods disclosed herein comprise one or more steps or actions for achieving the described method. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is specified, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

[0083] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A substrate processing method, comprising: chucking a substrate to a surface of a substrate support that is disposed in a processing volume, wherein chucking the substrate comprises: applying a first DC bias to a first pair of electrodes and a second pair of electrodes, the first pair of electrodes disposed within a first portion of the substrate support and the second pair of electrodes disposed within a second portion of the substrate support; detecting a difference in a first electrical characteristic of a first flow between the first pair of electrodes and the substrate and a second electrical characteristic of a second flow between the second pair of electrodes and the substrate; and applying a second DC bias to the first pair of electrodes and a third DC bias to the second pair of electrodes, wherein the second DC bias is different from the third DC bias, and the second DC bias and the third DC bias are selected based on detecting the difference in the first flow and the second flow.
2. The method of claim 1, wherein the second DC bias and the third DC bias are configured to reduce the difference in the first electrical characteristic and the second electrical characteristic.
3. The method of claim 1, wherein the first DC bias is configured to generate a first electrostatic force between the substrate and the first portion of the substrate support and the second portion of the substrate support.
4. The method of claim 3, wherein the second DC bias is configured to generate a second electrostatic force between the first portion of the substrate support and the substrate, the second electrostatic force having a greater magnitude than the first electrostatic force.
5. The method of claim 4, wherein the third DC bias is configured to generate a third electrostatic force between the second portion of the substrate support and the substrate.
6. The method of claim 1, wherein the difference in the first electrical characteristic and the second electrical characteristic corresponds to a non-uniform separation of a portion of the substrate from the surface of the substrate support, the portion of the substrate disposed between the first and second portions of the substrate support.
7. The method of claim 1, wherein chucking the substrate further comprises: detecting a change in the difference in the first electrical characteristic and the second electrical characteristic; and applying a fourth DC bias to the first pair of electrodes in response to detecting the change.
8. The method of claim 1, wherein second DC bias and the third DC bias are applied in a sequential order.
9. The method of claim 8, wherein the sequential order is configured to flatten the substrate before chucking the substrate.
10. The method of claim 8, wherein the sequential order is based on a magnitude of the second DC bias and a magnitude of the third DC bias.
11. The method of claim 8, wherein the sequential order is configured to minimize an amount of movement of the substrate relative to the substrate support.
12. The method of claim 8, wherein the sequential order is based on a portion of the substrate that initially contacts the surface of the substrate support.
13. The method of claim 8, wherein the sequential order is based on a determination that the substrate is at least one of concave or convex.
14. The method of claim 1, wherein chucking the substrate further comprises reducing the difference in the first electrical characteristic and the second electrical characteristic using a vacuum source in communication with a processing volume.
15. An apparatus, comprising: a substrate support; and a non-transitory computer readable medium storing executable instructions that, when executed by at least one processor, cause the at least one processor to perform operations for chucking a substrate to a surface of the substrate support, the operations comprising: applying a first DC bias to a first pair of electrodes and a second pair of

electrodes, the first pair of electrodes disposed within a first portion of the substrate support and the second pair of electrodes disposed within a second portion of the substrate support; applying an AC bias to the first pair of electrodes and the second pair of electrodes; detecting a difference in an electrical characteristic of a first electric flow flowing between the first pair of electrodes and a second electric flow flowing between the second pair of electrodes; and applying a second DC bias to the first pair of electrodes and a third DC bias to the second pair of electrodes, wherein the second DC bias is different from the third DC bias, and the second DC bias and the third DC bias are selected based on detecting the difference in the first electric flow and the second electric flow.

16. The apparatus of claim 15, wherein the second DC bias and the third DC bias are configured to reduce the difference in the first electric flow and the second electric flow, the first electric flow being a first AC current and the second electric flow being a second AC current.

17. The apparatus of claim 16, wherein the difference in the first AC current and the second AC current corresponds to a non-uniform separation of a portion of the substrate from the surface of the substrate support.

18. The apparatus of claim 17, wherein the first electrical characteristic is a first capacitance that corresponds to a first distance of the non-uniform separation and the second electrical characteristic is a second capacitance that corresponds to a second distance of the non-uniform separation.

19. The apparatus of claim 17, wherein the portion of the substrate is disposed between the first portion of the substrate support and the second portion of the substrate support.

20. The apparatus of claim 15, wherein the operations further comprise: detecting a change in the difference in the first electric flow and the second electric flow; and applying a fourth DC bias to the first pair of electrodes in response to detecting the change.

21. The apparatus of claim 15, wherein the first DC bias is configured to generate a first electrostatic force between the first and second portions of the substrate support and the substrate.

22. The apparatus of claim 21, wherein the second DC bias is configured to generate a second electrostatic force between the first portion of the substrate support and the substrate, the second electrostatic force having a greater magnitude than the first electrostatic force.

23. The apparatus of claim 22, wherein the third DC bias is configured to generate a third electrostatic force between the second portion of the substrate support and the substrate.

24. The apparatus of claim 15, wherein the difference in the first electrical characteristic and the second electrical characteristic is detected by measuring a first AC current associated with the first pair of electrodes and measuring a second AC current associated with the second pair of electrodes.

25. An electrostatic chuck (ESC) comprising: a substrate support; a plurality of pairs of bipolar electrodes that each have interdigitated conductors, a first plurality of bipolar electrodes of the plurality of pairs of bipolar electrodes in a first group of bipolar electrodes and a second plurality of bipolar electrodes of the plurality of pairs of bipolar electrodes in a second group of bipolar electrodes; and a plurality of chucking regions formed in a surface of the substrate support, wherein at least one pair of bipolar electrodes in the first plurality of bipolar electrodes and at least one pair of bipolar electrodes in the second plurality of bipolar electrodes are disposed within each of the plurality of chucking regions, and the first group of bipolar electrodes are configured to be separately biasable relative to the second group of bipolar electrodes.

26. The ESC of claim 25, wherein the surface of the substrate support has a first surface area and the plurality of chucking regions collectively have a second surface area and wherein the second surface area is about 10 percent or less of the first surface area.